

Introduction to Mathematics and Optimization

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The references in the suggested solutions to this examination paper are with respect to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (26.11.2020) of interactive book).

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = x^2 + y.$$

Also define the subset

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 1, y \leq 1, x + y \geq 1\}$$

of $\mathbb{R}^2$.

(a) Show that f is a convex function.

ANSWER:

The Hessian matrix of f is

$$\nabla^2(f) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

This is positive semidefinite according to Exercise 8.22. Therefore the function is convex by Theorem 8.17. Both references are important here. A reference to Exercise 3.40 is incorrect.

(b) Show that f is not strictly convex.

ANSWER:

The function cannot be strictly convex according to Definition 6.1, because $f(0, y) = y$. This implies that

$$f((1-t)(0, 0) + t(0, y)) = ty = (1-t)f(0, 0) + tf(0, y).$$

It is important here to go directly to Definition 6.1 and not fall into the trap of “using” Theorem 8.17.

Consider the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C. \end{aligned} \tag{1}$$

(c) Is the optimization problem (1) a convex optimization problem? If so, why?

ANSWER:

The optimization problem is convex, since f is a convex function, as shown earlier in the paper, and C is a convex set, being the intersection of the three convex sets $x \leq 1$, $y \leq 1$, and $1 - x - y \leq 0$. Here it is important to explain what this means and in that context refer to Exercise 4.18 and Exercise 4.15.

- (d) Express the set $U = \mathbb{R}^2 \setminus C$ using strict inequalities (i.e., $>$ and $<$) in x and y . Is U an open subset?

ANSWER:

Here we have

$$U = \{(x, y) \mid x > 1 \vee y > 1 \vee x + y < 1\}.$$

It is important here NOT to write

$$U = \{(x, y) \mid x > 1, y > 1, x + y < 1\}, \text{ which is interpreted as } \{(x, y) \mid x > 1 \wedge y > 1 \wedge x + y < 1\}.$$

This set is open, since the complement C is closed by 5.6.3. It is important here to go into detail about why C is closed with references and explanations.

- (e) Explain, by a simple reference to the curriculum without calculations, why there must exist an optimal solution to (1).

ANSWER:

The function f is continuous and C is closed and bounded for the same reasons as in the solutions to previous examination papers. Therefore an optimal solution exists by Theorem 5.51. Here there should be at least remarks about the set being bounded.

- (f) Solve the optimization problem (1) by **only** using the KKT conditions for (1). Is there more than one solution to the KKT conditions for (1)?

ANSWER:

The KKT conditions for (1) are according to (9.23)

$$\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0 \quad (2)$$

$$\lambda_1(x - 1) = 0 \quad (3)$$

$$\lambda_2(y - 1) = 0 \quad (4)$$

$$\lambda_3(1 - x - y) = 0 \quad (5)$$

$$2x + \lambda_1 - \lambda_3 = 0 \quad (6)$$

$$1 + \lambda_2 - \lambda_3 = 0 \quad (7)$$

If $\lambda_1 > 0$, then $x = 1$ by (3). We must then have $\lambda_3 > 0$ by (6). Hence $y = 0$ by (5), and therefore $\lambda_2 = 0$ by (4). Thus $\lambda_3 = 1$ and hence a contradiction with (6). We therefore have $\lambda_1 = 0$. If $\lambda_2 > 0$, then $y = 1$ by (4), and hence $\lambda_3 > 0$, and therefore $x = 0$, which gives $\lambda_1 = \lambda_3 > 0$. This is impossible since we must have $\lambda_1 = 0$. Hence we must have $\lambda_1 = \lambda_2 = 0$. From this we get $\lambda_3 = 1$ and $x = \frac{1}{2}$ and $y = \frac{1}{2}$. That is, a unique solution to the KKT conditions. Here it is important to argue for all cases to establish uniqueness. There should also be a reference to Theorem 9.26(ii).

PROBLEM 2

Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$g(x, y) = e^{x^2+y}.$$

(a) Show without using a computer that

$$\frac{\partial^2 g}{\partial x^2} = (2 + 4x^2)e^{x^2+y}.$$

ANSWER:

This follows by first differentiating once via the chain rule:

$$\frac{\partial g}{\partial x} = 2xe^{x^2+y}$$

and then differentiating once more via the product rule and chain rule:

$$\frac{\partial^2 g}{\partial x^2} = 2e^{x^2+y} + 4x^2e^{x^2+y} = (2 + 4x^2)e^{x^2+y}.$$

It is important here to state which differentiation rules are being used precisely to show that no computer was used and that the procedure is understood.

(b) Show that a symmetric 2×2 matrix

$$\begin{pmatrix} a & c \\ c & b \end{pmatrix}$$

with $a, b, c \in \mathbb{R}$ is positive definite if and only if $a > 0$ and $ab - c^2 > 0$.

ANSWER:

This is the content of Exercise 3.40. A reference is sufficient here.

(c) Show that g is a strictly convex function.

ANSWER:

To use Theorem 8.17, we find the partial derivatives:

$$\begin{aligned} \frac{\partial g}{\partial x} &= 2xe^{x^2+y} \\ \frac{\partial g}{\partial y} &= e^{x^2+y} \\ \frac{\partial^2 g}{\partial x^2} &= 2e^{x^2+y} + 4x^2e^{x^2+y} \\ \frac{\partial^2 g}{\partial y^2} &= e^{x^2+y} \\ \frac{\partial^2 g}{\partial x \partial y} &= 2xe^{x^2+y} \end{aligned}$$

This gives

$$\nabla^2(g) = \begin{pmatrix} 2e^{x^2+y} + 4x^2e^{x^2+y} & 2xe^{x^2+y} \\ 2xe^{x^2+y} & e^{x^2+y} \end{pmatrix}$$

From the previous part and Theorem 8.17, we see that f is strictly convex, since

$$(2 + 4x^2)e^{x^2+y} > 0$$

$$(2 + 4x^2)e^{x^2+y}e^{x^2+y} - 4x^2e^{x^2+y}e^{x^2+y} = 2(e^{x^2+y})^2 > 0$$

for all $x, y \in \mathbb{R}$. Here the calculations and the connection to the previous part are important. One should also note why the above inequalities hold, among other things by noting that the exponential function is always positive.

(d) Does the function $g(x, y)$ have a global minimum?

ANSWER:

A global minimum (x_0, y_0) would, according to Proposition 7.16, have to satisfy

$$\frac{\partial g}{\partial y}(x_0, y_0) = e^{x_0^2+y_0} = 0,$$

which is impossible. This reference is important.

Now let C denote the subset of $\mathbb{R}^2$ given by (as in PROBLEM 1)

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 1, y \leq 1, x + y \geq 1\}$$

and consider the optimization problem

$$\begin{aligned} \min g(x, y) \\ (x, y) \in C. \end{aligned} \tag{8}$$

(e) Why does the optimization problem (8) have a unique solution?

ANSWER:

We know, exactly as in Problem 1(e), that an optimal solution exists. This solution is unique by Theorem 6.13, since g is strictly convex.

(f) Solve the optimization problem (8).

ANSWER:

The function e^z is increasing, so we only need to minimize $x^2 + y$. This was done in Problem 1. That is, it is the same optimum as in Problem 1(e), i.e., $x = y = \frac{1}{2}$.

(g) Does the optimization problem (8) have a unique solution if we replace

$$g(x, y) = e^{x^2+y}$$

with

$$g(x, y) = e^{x+y}?$$

ANSWER:

Since we have $x + y \geq 1$, we must have $e^{x+y} \geq e^1 = e$ when $(x, y) \in C$. But all points $(x, y) \in C$ with $x + y = 1$ are then optima. That is, the optimum is not unique.

PROBLEM 3

Consider the system of linear equations

$$\begin{aligned}x + y + z &= 1 \\x + 2y + 3z &= 1 \\y + 2z &= 1\end{aligned}\tag{9}$$

in the three variables x, y , and z .

(a) Let

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Find a 3×3 matrix A and a 3×1 matrix b such that the system of equations (9) is expressed as $Av = b$.

ANSWER:

By the definition of matrix multiplication,

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

can be used.

(b) Show that there exists no $v \in \mathbb{R}^3$ such that $Av = b$ (the system of equations (9) cannot be solved).

ANSWER:

When the first equation is subtracted from the second equation, we obtain the equation $y + 2z = 0$, which contradicts the last equation $y + 2z = 1$.

(c) Show that the matrix A is not invertible.

ANSWER:

If the matrix A were invertible, then $v = A^{-1}b$ would satisfy $Av = b$. Therefore A cannot be invertible by the previous part. It is important here to base the argument on the curriculum to solve the problem.

(d) Solve the optimization problem

$$\min_{v \in \mathbb{R}^3} |b - Av|^2.$$

It is permitted to use a computer for matrix calculations here. Is the solution to the optimization problem unique?

ANSWER:

Here we use Theorem 5.11 and must solve the system of equations

$$(A^T A)v = A^T b.\tag{10}$$

Here

$$A^T A = \begin{pmatrix} 2 & 3 & 4 \\ 3 & 6 & 9 \\ 4 & 9 & 14 \end{pmatrix} \quad \text{and} \quad A^T b = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}.$$

It is verified by computer that (10) has the solution

$$x = y = z = \frac{2}{9}.$$

There are many solutions to this optimization problem. For the vector

$$u = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

we have $Au = 0$. If v is a solution to the optimization problem, $v + u$ will also be one, because

$$|b - Av|^2 = |b - A(v + u)|^2.$$

One can also argue from a complete solution to $(A^T A)v = A^T b$ and see that it depends on a variable z :

$$x = z, \quad y = \frac{2}{3} - 2z, \quad z = z.$$